

Adversarial Cloak – Project Report

1. Project Overview

Adv_cloak is a Python-based research pipeline that implements adversarial patch attacks against the YOLOv2 person detection model. The system generates, trains and applies adversarial patches designed to suppress object detection confidence scores, effectively making person invisible to the detector.

2. System Architecture

The pipeline consists of three sequentially executed phases

Phase	Script	Description
Train	train_yolo_patch.py	Optimises a 400x400 adversarial patch tensor via gradient descent (Adam, lr = 0.01) over frozen YOLOv2 weights. Loss = objectness suppression + Total Variation regularisation.
Apply	apply_patch	Detects people with YOLOv2, positions patch on the torso region using face-aware placement and composites result to disk.
Detect	detect_people.py	Detects people on provided images with YOLOv2 model and saves the resulting images to disk.
Real-Time	real_time_detection.py	Runs detection script on video from user camera.

3. Patch Training

Loss Function

The total loss combines to components:

- Detection loss (objectness): mean objectness score across all YOLOv2 anchor outputs
- Total Variation loss: penalizes pixel-level noise to keep the patch printable and visually coherent. (tv_weight = 2.0 at first try, but was raised to 4.0 for smoother output).

Key hyperparameters

Parameter	Value	Notes
patch_size	400px	Square RGB tensor, clamped to [0, 1]
img_size	416px	YOLOv2 native input resolution
epochs	50 -> 200	Recommened: 200-300
batch_size	4	Increase if VRAM allows
lr	0.01	Adam
tv_weight	2.0 -> 4.0	Raised to 4.0 for smoother output
patch scale	20-30%	Random scale applied per batch

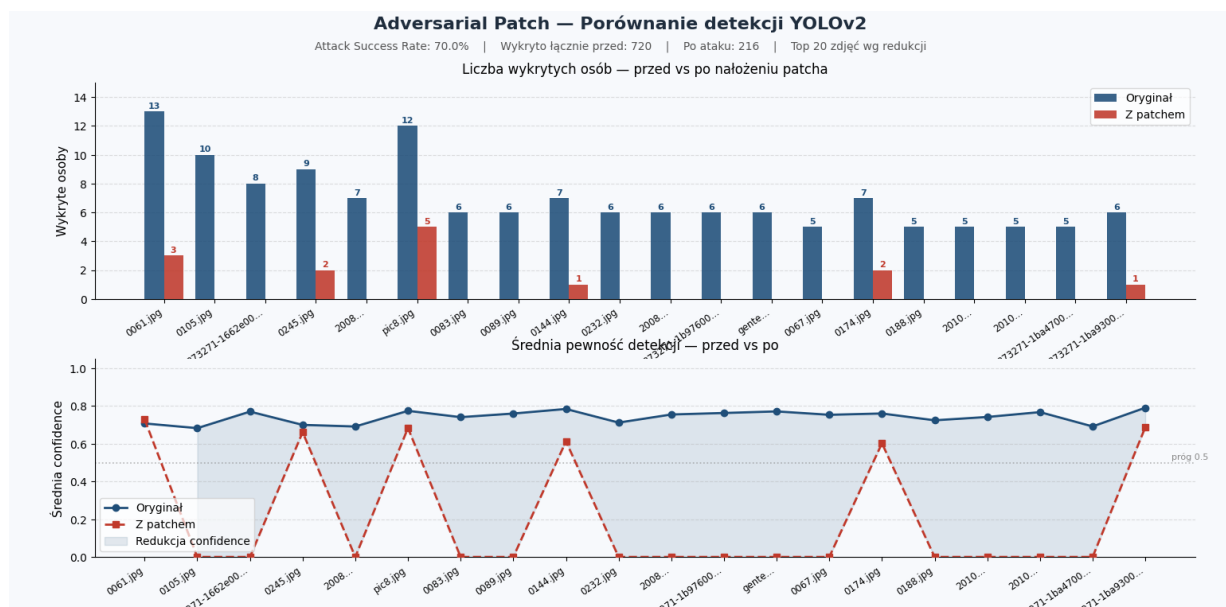
4. Patch Placement Strategy

Patch placement uses a two-tier approach to avoid covering the face:

- Primary: face detector is run on the person's bounding box, if a face is detected, the patch is placed below it with a 20% margin.
- Fallback: When no face detected (profile views) the patch is placed at 22-62% of the bounding box height – corresponding to the chest region.
- Patch dimensions: 55% of bbox width, 80% of region height, horizontally centred on the bbox.

5. Attack Effectiveness

The adversarial patch was evaluated on 260 images using YOLOv2. The chart shows the 20 cases with the highest detection reduction. Overall Attack Success Rate reached **70.0%**, suppressing 504 out of 720 original detections. Confidence scores dropped from a stable 0.65–0.80 range to near 0.0 in most patched cases. Results were achieved after 200 training epochs.



6. Limitations and Improvements

- 50 training epochs wasn't enough for a strong patch – 200 or more is recommended
- Tv_weight = 2.0 produced visually noisy patches, raising to 4.0 improved printability at a small cost to attack strength.
- Patch placement is fixed-size and centred; a physically realistic implementation should use perspective transform and person pose estimation
- YOLOv2 is an older architecture, newer detectors could perform better.